

Chengzu Li

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EDUCATION

Language Technology Lab, University of Cambridge

Ph.D. in Computation, Cognition and Language

Cambridge, UK

2023 - current

- Supervised by Dr. Ivan Vulić, Prof. Anna Korhonen and Prof. Serge Belongie, fully funded by Cambridge Trust.
- Visiting student at Belongie Lab (2025.8 - 2025.11)

University of Cambridge

M.Phil. in Advanced Computer Science; Graduated with Distinction (81.7/100)

Cambridge, UK

2022 - 2023

- Supervised by Prof. Simone Teufel, fully funded by Cambridge Trust.
- M.Phil. thesis: Generating Instructions for Grounded Robot Navigation (*Distinction*)

Xi'an Jiaotong University

B.Eng. in Automation, minor in Fintech; GPA: 91.96/100 (top 5%)

Xi'an, China

2018 - 2022

University of Hong Kong

Exchange student in Computer Engineering; GPA: 4.08/4.3

Student Research Assistant in Dept. of Computer Science

Hong Kong

Jan. 2021 - Jun. 2021

Jul. 2021 - Sept. 2021

SELECTED PUBLICATIONS & PROJECTS

Imagine While Reasoning in Space: Multimodal Visualization-of-Thought

ICML 2025

Chengzu Li*, Wenshan Wu*, Huanyu Zhang, Yan Xia, Shaoguang Mao, Li Dong, Ivan Vulić, Furu Wei

- *Media Engagement*: Featured at [IEEE Spectrum](#), [TWIML AI Podcast](#) and [etc.\(CHN\)](#)

- *Presentation*: CHIA 2025. [Invited Talk at HKUST\(GZ\)](#)

TopViewRS: Vision-Language Models as Top-View Spatial Reasoners

EMNLP 2024 **Oral**

Chengzu Li*, Caiqi Zhang*, Han Zhou, Nigel Collier, Anna Korhonen, Ivan Vulić

- *Media Engagement*: Featured at [TWIML AI Podcast](#)

- *Presentation*: Non-archival at [ACL 2023 SpLU Robo Workshop \(Oral\)](#), [ECCV 2024 Eval-Fomo Workshop](#)

Visual Planning: Let's Think Only with Images

arXiv Preprint

Yi Xu*, Chengzu Li*, Han Zhou*, Xingchen Wan, Caiqi Zhang, Anna Korhonen, Ivan Vulić

- *Media Engagement*: Featured at [X\(Twitter\)](#), [Medium](#) and [etc.\(CHN\)](#)

- *Community Influence* with over 250 Github stars

- *Presentation*: Non-archival at [Foundation Models Meet Embodied Agents@CVPR 2025](#); [CHIA 2025](#)

PROFESSIONAL EXPERIENCES

GenAI Group, Microsoft Research, Research Intern

Jun. 2024 - Jan. 2025

Toshiba Cambridge Research Laboratory, Research Intern

Jan. 2023 - Jul. 2023

NLP Group, Shanghai Artificial Intelligence Laboratory, Research Intern

Oct. 2021 - Jan. 2022

University of Hong Kong CS Research Internship Programme, Research Intern

Jul. 2021 - Oct. 2021

FULL PUBLICATIONS

Information Loss in Vision-Language Models

EMNLP 2025 Findings

Wenyan Li, Raphael Tang, Chengzu Li, Caiqi Zhang, Ivan Vulić, Anders Søgaard

Enriching Patent Claim Generation with European Patent Dataset

EMNLP 2025 Findings

Lekang Jiang, Chengzu Li, Stefan Goetz

Imagine While Reasoning in Space: Multimodal Visualization-of-Thought

ICML 2025

Chengzu Li*, Wenshan Wu*, Huanyu Zhang, Yan Xia, Shaoguang Mao, Li Dong, Ivan Vulić, Furu Wei

Large Language Models are Miscalibrated In-Context Learners

ACL 2025 Findings

Chengzu Li, Han Zhou, Goran Glavaš, Anna Korhonen, Ivan Vulić
- *Oral Presentation at BMVA Workshop, Non-archival at ICLR 2024 Workshop on R2FM*

TopViewRS: Vision-Language Models as Top-View Spatial Reasoners EMNLP 2024 **Oral**

Chengzu Li*, Caiqi Zhang*, Han Zhou, Nigel Collier, Anna Korhonen, Ivan Vulić

Semantic Map-based Generation of Navigation Instructions COLING 2024

Chengzu Li, Chao Zhang, Simone Teufel, Rama Sanand Doddipatla, Svetlana Stoyanchev

Generating Data for Symbolic Language with Large Language Models EMNLP 2023 Main

Jiacheng Ye, Chengzu Li, Lingpeng Kong, Tao Yu

Binding Language Models in Symbolic Languages ICLR 2023 **Spotlight**

Zhoujun Cheng*, Tianbao Xie*, Peng Shi, Chengzu Li, Rahul Nadkarni, Yushi Hu, Caiming Xiong, Dragomir Radev, Mari Ostendorf, Luke Zettlemoyer, Noah A. Smith, Tao Yu

UnifiedSKG: Unifying and Multi-Tasking Structured Knowledge Grounding with Text-to-Text Language Models EMNLP 2022 **Oral**

Tianbao Xie*, Chen Henry Wu*, Peng Shi, Ruiqi Zhong, Torsten Scholak, Michihiro Yasunaga, Chien-Sheng Wu, Ming Zhong, Pengcheng Yin, Sida I. Wang, Victor Zhong, Bailin Wang, Chengzu Li, Connor Boyle, Ansong Ni, Ziyu Yao, Dragomir Radev, Caiming Xiong, Lingpeng Kong, Rui Zhang, Noah A. Smith, Luke Zettlemoyer, Tao Yu
(*: equal contribution)

PRE-PRINTS

11Plus-Bench: Demystifying Multimodal LLM Spatial Reasoning with Cognitive-Inspired Analysis

Chengzu Li, Wenshan Wu, Huanyu Zhang, Qingtao Li, Zeyu Gao, Yan Xia, José Hernández-Orallo, Ivan Vulić, Furu Wei

Visual Planning: Let's Think Only with Images

Yi Xu*, Chengzu Li*, Han Zhou*, Xingchen Wan, Caiqi Zhang, Anna Korhonen, Ivan Vulić

Scaling and Beyond: Advancing Spatial Reasoning in MLLMs Requires New Recipes

Huanyu Zhang*, Chengzu Li*, Wenshan Wu, Shaoguang Mao, Yifan Zhang, Haochen Tian, Ivan Vulić, Zhang Zhang, Liang Wang, Tieniu Tan, Furu Wei

Reinforcement Learning for Better Verbalized Confidence in Long-Form Generation

Caiqi Zhang*, Xiaochen Zhu*, Chengzu Li, Nigel Collier, Andreas Vlachos

TEACHING EXPERIENCES

Teaching Assistant	Department of Theoretical and Applied Linguistics, University of Cambridge
Li18 Computational Linguistics (<i>Practical Session</i>)	2023,2024

SELECTED AWARDS & GRANTS

2025	Postgraduate Research Fund from Jesus College, Cambridge
2025	Lambda Research Grant (\$2000 USD in GPU credits)
2024	Postgraduate Research Fund from Jesus College, Cambridge
2024	Scholar of Jesus College, Cambridge (for outstanding academic performance)
2023	PhD Scholarship from Cambridge Trust (Fully-Funded)
2022	Masters Scholarship from Cambridge Trust (Fully-Funded)
2019	National Scholarship (Top 1%, highest honor for undergraduates in China)
2019	Excellent Student (Top 10% among 5000+ students)

MEDIA COVERAGE & PRESENTATIONS

2025	Invited Talk at HKUST (GZ): Reason with Multimodal Minds in Space
2025	LiangZiWei (CHN): Imagined Visuals Alone Boost Large Model Reasoning Accuracy by 80%
2025	JiQiZhiXin (CHN): Thinking with Images Alone: RL Fosters a New Paradigm for Reasoning Models
2025	XinZhiYuan (CHN): Beyond CoT! Directly Visualize the Multimodal Reasoning Process
2025	TWIML AI Podcast with Sam Charrington: Multimodal Visualization-of-Thought
2025	IEEE Spectrum, "Thinking" Visually Boosts AI Problem Solving
2024	BMVA: Trustworthy Multimodal Learning with Foundation Models